DATA WAREHOUSING AND MINING LABORATORY

Experiment No. 5: Generate Association Rules Using Apriori Algorithm

Aim: To generate frequent itemsets and association rules using the Apriori algorithm in WEKA and analyze the relationships among items in transactional datasets.

Objectives: After successful completion of this experiment, students will be able to:

1. Understand the concept of association rule mining.
2. Understand frequent itemsets.
3. Calculate and interpret support and confidence.
4. Apply the Apriori algorithm using WEKA.
5. Generate association rules from transactional data.
6. Analyze discovered patterns.
7. Interpret association rules for decision making.

Importance of this Experiment:

Large organizations collect millions of transaction records every day. Association rule mining helps discover hidden relationships among items and supports:

- Product recommendation
- Market basket analysis
- Customer behavior analysis
- Food recommendation systems
- Cross-selling strategies

The Apriori algorithm is one of the most widely used approaches for discovering such relationships.

Expected Outcome: After completing this experiment, students will be able to:

1. Generate frequent itemsets.
2. Generate association rules.
3. Interpret support and confidence values.
4. Analyze customer purchase behavior.
5. Apply association rule mining concepts to recommendation systems.

Facilities Required:

Hardware Requirements:

- Personal Computer / Laptop
- Intel i3 Processor or higher
- Minimum 4 GB RAM
- Internet Connection (Optional)

Software Requirements:

- Windows 10/11
- WEKA Tool
- Market Basket Dataset (market_basket.arff)

Prerequisite Knowledge: Students should be familiar with:

- Experiment No. 1: Installation and Exploration of WEKA Tool and Available Datasets
- Experiment No. 2: Create and Analyze ARFF Files Using WEKA
- Experiment No. 3: Data Cleaning and Missing Value Treatment
- Experiment No. 4: Data Transformation and Normalization
- Basic Data Mining Concepts
- Attributes and Instances

Theory:

1. Frequent Pattern Mining

Frequent pattern mining is the process of discovering recurring relationships among data items. For example, Milk, Bread appearing together in many transactions. Such recurring patterns are called frequent patterns.

2. Association Rule Mining

Association Rule Mining is a data mining technique used to discover relationships among items in large datasets.

General form: A → B

Meaning: If item A occurs, item B is likely to occur.

Example: Milk → Bread

Interpretation: Customers purchasing milk often purchase bread.

3. Frequent Itemsets

A Frequent Itemset is a set of items that appears together frequently in transactions.

Examples:

{Milk}

{Milk, Bread}

{Milk, Bread, Eggs}

Frequent itemsets form the basis for generating association rules.

4. Support

Support measures how frequently an itemset appears in the dataset.

Formula:

Support (A) = Transactions Containing A / Total Transactions

Example:

If Milk appears in 4 out of 5 transactions:

Support (Milk) = 4/5 = 80%

Higher support indicates that the item occurs frequently.

5. Confidence

Confidence measures the reliability of an association rule.

Formula:

Confidence (A→B) = Transactions Containing A and B / Transactions Containing A

Example:

If 4 customers buy Milk and 3 of them also buy Bread:

Confidence (Milk→Bread) = 3/4 = 75%

Higher confidence indicates a stronger relationship.

6. Apriori Algorithm

The Apriori algorithm is used to:

- Discover frequent itemsets
- Generate association rules
- Eliminate infrequent combinations

Apriori Principle: **If an itemset is frequent, all of its subsets must also be frequent.**

Example:

If {Milk, Bread, Eggs} is frequent, then {Milk, Bread} must also be frequent.

Industry Insight: Modern retailers, supermarkets, e-commerce platforms, and recommendation systems use association rule mining to understand purchasing behavior and recommend products that are likely to be purchased together.

Mini Real-World Example:

Supermarket Shopping Analysis

Transaction ID	Items Purchased
T1	Milk, Bread
T2	Milk, Eggs
T3	Bread, Butter
T4	Milk, Bread, Eggs
T5	Milk, Bread, Butter

Possible Pattern: Milk → Bread

This suggests that customers buying Milk tend to purchase Bread as well.

Food Recommendation System

Child	Preferred Foods
C1	Milk, Banana
C2	Milk, Apple
C3	Milk, Banana
C4	Banana, Apple

Possible Rule: Milk → Banana

Such rules can assist in generating food recommendations.

Dataset Used in This Experiment:

Filename: market_basket.arff

Dataset Contents:

@relation marketbasket

@attribute Milk {t,f}

@attribute Bread {t,f}

@attribute Eggs {t,f}

@attribute Butter {t,f}

@data

t,t,f,f

t,f,t,f

f,t,f,t

t,t,t,f

t,t,f,t

Dataset Information:

- Relation Name: marketbasket
- Number of Attributes: 4
- Number of Transactions: 5
- Attribute Type: Nominal

Before Applying Apriori:

Sample Transactions

Transaction ID	Items Purchased
T1	Milk, Bread
T2	Milk, Eggs
T3	Bread, Butter
T4	Milk, Bread, Eggs
T5	Milk, Bread, Butter

Predict the Following

Before running Apriori, try identifying possible relationships:

Milk → Bread

Bread → Butter

Milk → Eggs

After executing the algorithm, compare your predictions with the generated rules.

Understanding Generated Rules:

Example Rule: Milk=t ==> Bread=t

Interpretation: Customers purchasing Milk are likely to purchase Bread.

Support Interpretation

Example: Support = 60%

Meaning: The rule appears in 60% of all transactions.

Confidence Interpretation

Example: Confidence = 80%

Meaning: 80% of customers purchasing Milk also purchase Bread.

Procedure:

Activity 1: Load Dataset

Steps:

1. Open WEKA
2. Select Explorer
3. Open the Preprocess tab
4. Click Open File
5. Select market_basket.arff
6. Click Open

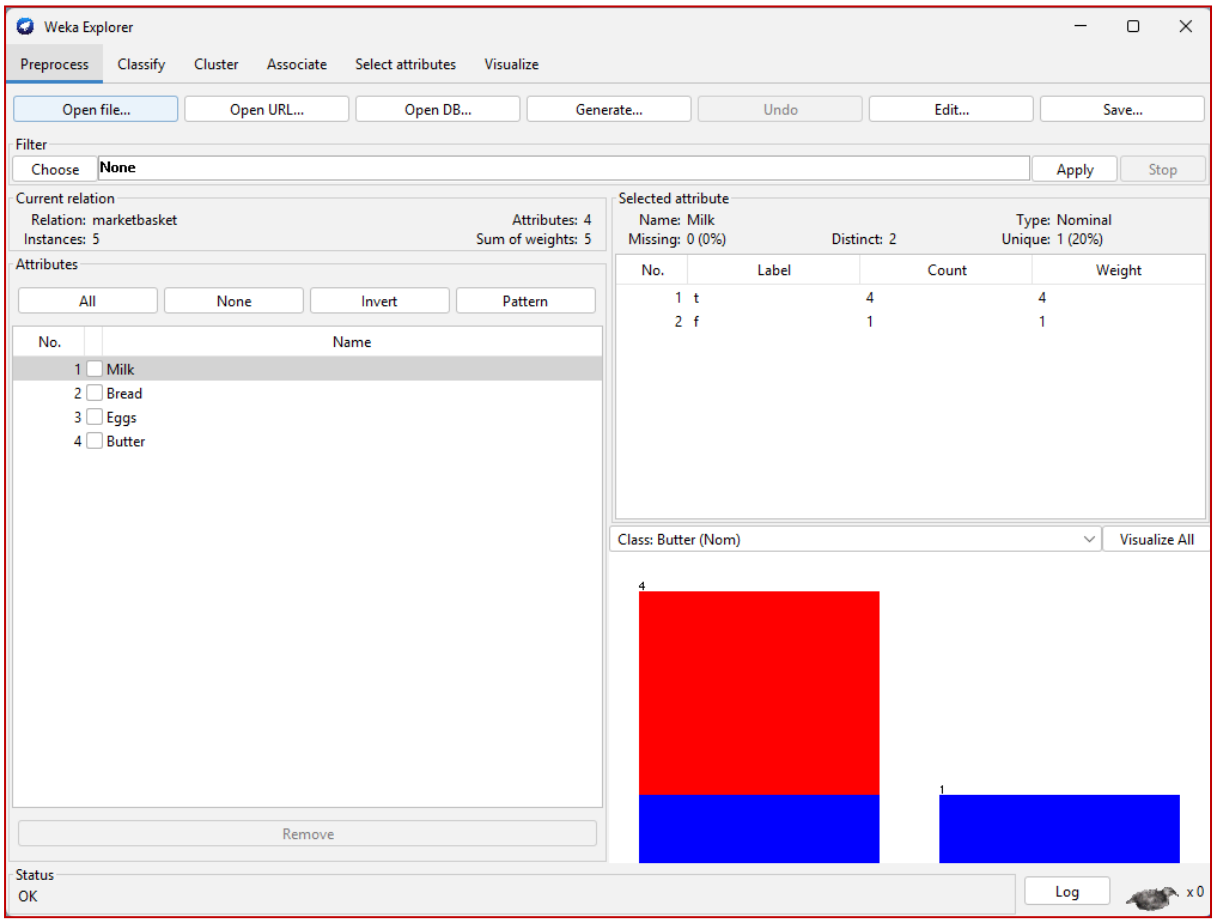


Figure 5.1: Market Basket Dataset Loaded in WEKA

Activity 2: Open Associate Tab

Step: Select: Associate tab.

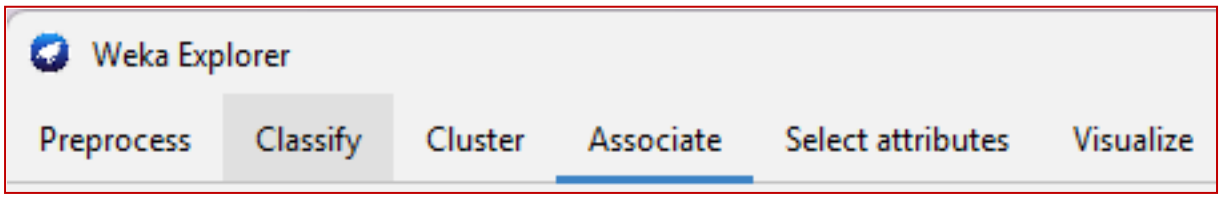


Figure 5.2: WEKA Associate Tab

Activity 3: Select Apriori Algorithm

Steps:

1. Click: Choose
2. Select: Apriori

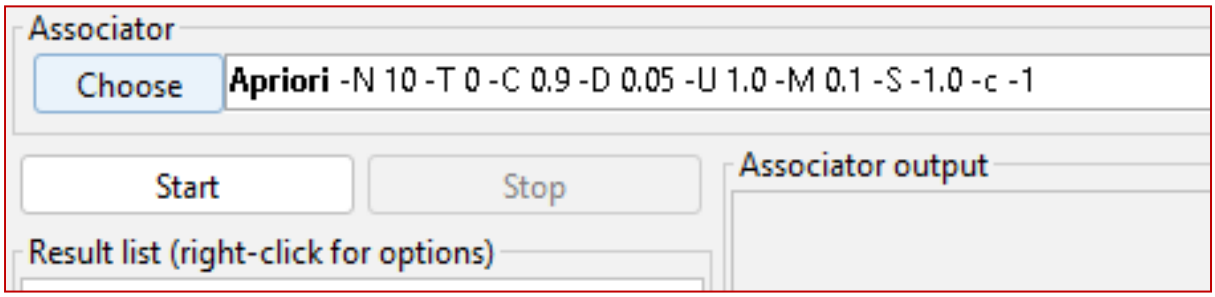


Figure 5.3: Selecting Apriori Algorithm

Activity 4: Generate Association Rules

Steps:

1. Click: Start
2. Observe generated:
   - Frequent Itemsets
   - Support Values
   - Confidence Values
   - Association Rules

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=== Run information ===

Scheme:       weka.associations.Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1
Relation:     marketbasket
Instances:    5
Attributes:   4
              Milk
              Bread
              Eggs
              Butter
=== Associator model (full training set) ===


Apriori
=======

Minimum support: 0.5 (2 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 10

Generated sets of large itemsets:

Size of set of large itemsets L(1): 6

Size of set of large itemsets L(2): 9

Size of set of large itemsets L(3): 4

Best rules found:

 1. Butter=f 3 ==> Milk=t 3    <conf:(1)> lift:(1.25) lev:(0.12) [0] conv:(0.6)
 2. Eggs=f 3 ==> Bread=t 3    <conf:(1)> lift:(1.25) lev:(0.12) [0] conv:(0.6)
 3. Eggs=t 2 ==> Milk=t 2    <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.4)
 4. Butter=t 2 ==> Bread=t 2    <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.4)
 5. Eggs=t 2 ==> Butter=f 2    <conf:(1)> lift:(1.67) lev:(0.16) [0] conv:(0.8)
 6. Butter=t 2 ==> Eggs=f 2    <conf:(1)> lift:(1.67) lev:(0.16) [0] conv:(0.8)
 7. Milk=t Eggs=f 2 ==> Bread=t 2    <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.4)
 8. Bread=t Butter=f 2 ==> Milk=t 2    <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.4)
 9. Eggs=t Butter=f 2 ==> Milk=t 2    <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.4)
10. Milk=t Eggs=t 2 ==> Butter=f 2    <conf:(1)> lift:(1.67) lev:(0.16) [0] conv:(0.8)
```

Figure 5.4: Generated Association Rules Using Apriori

Activity 5: Interpret the Results

Identify:

- Strongest Rule
- Highest Confidence Rule
- Most Frequent Itemset

Record your observations.

Common Mistakes While Applying Apriori

1. Confusing support with confidence.
2. Assuming every generated rule is important.
3. Ignoring confidence values.
4. Using unsuitable datasets.
5. Generating rules without interpreting their business meaning.

Common Discussion

1. What is association rule mining?
2. What is a frequent itemset?
3. What is support?
4. What is confidence?
5. Why is Apriori useful?
6. What industries use association rules?
7. How can association rules support recommendation systems?

Student Activity:

Create a market basket dataset having:

- At least 5 items
- At least 10 transactions

Apply Apriori and identify:

1. Strongest Rule
2. Rule with Highest Confidence
3. Most Frequent Itemset

Modify:

- Minimum Confidence
- Number Of Rules To Be Generated

Observe:

- Number of rules generated

- Strongest rule generated
- Changes in support and confidence values

Record your findings.

Observation Table:

PARAMETER	OBSERVATION
Dataset Name	
Dataset File Name	
Number of Attributes	
Number of Transactions	
Algorithm Used	
Minimum Support	
Minimum Confidence	
Total Rules Generated	
Strongest Rule Generated	
Highest Confidence Value	

Result: The Apriori algorithm was successfully applied to the transactional dataset using WEKA. Frequent itemsets and association rules were generated and analyzed. The concepts of support, confidence, and association rule mining were studied successfully.

Viva Questions:

1. What is association rule mining?
2. What is a frequent itemset?
3. Define support.
4. Define confidence.
5. What is Apriori?
6. Explain the Apriori principle.
7. What is market basket analysis?
8. How are association rules represented?

9. Why is support important?
10. Why is confidence important?
11. Give two applications of association rule mining.
12. What is the difference between frequent itemsets and association rules?
13. Can Apriori be used in recommendation systems?
14. Which WEKA tab is used for Apriori?
15. What output does Apriori generate?

Assignment Questions:

1. Explain the Apriori algorithm with a suitable example.
2. Differentiate between support and confidence.
3. Explain market basket analysis.
4. Create a transactional dataset and generate association rules using WEKA.
5. Explain applications of association rule mining in recommendation systems.
6. Explain the Apriori principle with an example.
7. Compare frequent itemsets and association rules.

References:

1. Ian H. Witten, Eibe Frank and Mark A. Hall, *Data Mining: Practical Machine Learning Tools and Techniques*, Morgan Kaufmann.
2. Jiawei Han, Micheline Kamber and Jian Pei, *Data Mining: Concepts and Techniques*, Morgan Kaufmann.
3. WEKA Official Documentation & User Manual.

Prof. F I Khandwani	**Dr. S D Padiya**
Subject In-charge	**Head of the Department**